

Babäk Firoozi Fooladi

Location: Espoo, Uusimaa, Finland
LinkedIn | GitHub | StackOverFlow
Email: b.firoozi.f@gmail.com

Data Analyst | Data Engineer

Applied Data Scientist specializing in econometric modeling, time-series, and AI-driven data extraction using local LLMs. Combines rigorous paper-to-code research capabilities with production tooling (Python, R, Docker, Git).



Core Competencies & Technical Skills

- **Position related skills:** Data Cleaning, Data validation, research data, Quantitative research, spatial econometrics, statistics, LLM feature extraction, RAG.
- **Data Analytics & BI:** Power BI, Dashboard Development, Data Visualization, Quantitative Research, Business Intelligence, Technical Documentation
- **Programming & Core Languages:** Python, R, SQL, Julia, STATA, Git
- **Data Engineering & Databases:** PostgreSQL, PostGIS, DuckDB, Pandas, Polars, GeoParquet, SQLite, MongoDB, REST APIs, Web Scraping, ETL/ELT Pipelines, Azure Functions, GCP, Azure
- **Spatial Data & GIS Analytics:** GeoPandas, Shapely, ArcGIS / ArcGIS Online, ESRI CityEngine, Spatial Econometrics, Cartography, Network Analysis
- **Data Science & ML:** Machine Learning, Econometric Modeling, Time-Series Forecasting (VAR, Co-integration, Univariate/Multivariate), OLS / IV-OLS, Panel Data, Survival Analysis, RAG, Agentic AI, LLMs/NLP

Professional Experience

DATA ANALYTICS & ENGINEERING <i>Self-Employed Consultant</i>	Jan 2025 – Present <i>Project-Based – Espoo, Finland</i>
• Deliver end-to-end data engineering, spatial analytics, and business intelligence solutions tailored to client objectives and scalable growth. • Backend Architecture & Spatial Engineering (Apex-Heat): Architected automated Python scripts using GeoPandas, Shapely, GeoParquet, and PostGIS to acquire, clean, geometry-correct, and model national spatial building datasets across Finland. • API & Data Pipeline Automation: Designed and implemented API-driven data pipelines for automated spatial operations, evaluating data models for EU-wide scaling using domain expertise in real estate economics and zoning laws. • Pipeline Optimization & Tech Debt Reduction (Apex-Heat): Spearheaded core codebase migration from Pandas to Polars, refactoring design patterns and introducing comprehensive test coverage to achieve an 86% performance improvement on targeted calculations and a 60% overall application speedup.	
Ph.D. Candidate <i>Aalto University</i>	Sep 2020 – June 2026 <i>Full-time – Espoo, Finland</i>
• Conduct quantitative research at the intersection of urban economics and real estate, applying econometrics and machine learning to housing market dynamics and spatial development. • Architected and managed team data infrastructure, deploying a partitioned Data Lake and DuckDB-based querying workflows with full technical documentation for non-technical researchers.	

- Engineered automated web scraping pipelines using Azure Functions and REST/ArcGIS endpoints to harvest, geocode, and store real-world property transaction data (MML, Ryhti, SYKE, KVKL).
- Modeled complex urban dynamics using time-series analysis, hedonic regressions, and time-to-event statistical modeling.
- Applied NLP and RAG architectures to process PDF/HTML land-use planning documents for structured feature extraction.

Research Assistant

Jun 2019 – Sep 2019

Aalto University Full-time – Espoo, Finland

- Collaborated with **MIT CityLab** on Agent-Based Modeling (ABM) to simulate urban mobility and parking pressure dynamics across campus infrastructure using the GAMA Platform.

Data Analytics Engineer

Sep 2020 – Dec 2024

Qissa kaupunkisuunnitteluanalytiikka Oy – Startup

Part-time – Espoo, Finland

- Designed and maintained end-to-end spatial data pipelines ingesting REST API data, performing geocoding and automated storage for urban planning analytics.
- Developed a spatial analytics dashboard with a custom network analysis backend, integrating HSL, HERE, and HSY APIs for real-time transportation routing and network evaluation.
- Processed multi-national census datasets (US, Finland, Sweden, Denmark) to calculate commuting pattern carbon emissions.

Planning Assistant

Nov 2018 – Jun 2021

WSP Finland Oy

Part-time – Helsinki, Finland

- Performed spatial data analysis and pedestrian network modeling to forecast route choices and post-development pedestrian traffic for municipal clients.
- Built interactive 3D city models using ESRI CityEngine and configured ArcGIS Online platforms for maritime spatial planning projects.
- Produced BI reports, spatial maps, and visualizations to communicate quantitative findings to multidisciplinary teams and executive stakeholders.

GIS Operator

Aug 2015 – May 2016

University of Tehran

Part-time – Tehran, Iran

- Collected, processed, and prepared **GIS datasets** to support urban designers in implementing and evaluating new urban design directives for Tehran.

Advisor & Assistant

Aug 2011 – Aug 2018

Airsa Dorsa Co. Ltd. – Family Business

Full-time – Karaj, Iran

- Managed financial planning, cost metrics, and operational workflows, leveraging accounting datasets to generate business intelligence reporting for executive decision-making.
- Double booking accounting, cost calculation, document management, and financial reporting for manufacturing operations.

Education

Aalto University

Doctor of Philosophy in Technology, Real Estate Economics

Espoo, Finland

Sep 2020 - Present

Aalto University

MSc. (Tech.) Urban Studies and Planning in Real Estate Economic

GPA: 4.1/5, Graduated with honours

Espoo, Finland

Sep 2018 - Jul 2020

languages

- Persian (Native)
- English (Fluent (C2))
- Finnish (Intermediate (A2-B1))

Publications:

- Effects of zoning on property values and impact assessment of changes in land use policy measures: [Source](#)

Certificates:

- Oxford Machine Learning Summer School, OxML 2023
- Urban Economic Association, UEA 2020 Lectures on Urban Economics

Community & Leadership

- **Speaker:** Spatial data analysis, spatial data engineering. DataTribe collective meetup, 2023
- **Contributor:** Helsinki Data Week volunteer (2024 & 2025)
- **Speaker:** "The Effect of Land Development Policies on Implementation of Housing Plans". 28th Annual European Real Estate Society conference, 22nd-25th June, Milan, Italy.
- **Speaker:** "The Impact of Public Land Development on Implementation of Housing Plans". AREUEA International Conference, 19th-22nd July, Cambridge, UK.
- **Speaker:** "Time limit: is it an effective way to encourage housing construction?". RYTS Science Slam, 25th of January, Helsinki, Finland
- **Speaker:** "Balancing financial constraints and public objectives: Assessing the impact of public land ownership on housing development duration". World Planning School Congress, 17th-21st July, Espoo, Finland
- DataTribe collective moderator
- **Guest Lecturer:** Science and Breakfast Staff Meeting, Introduction to Zotero, how to migrate from Mendeley. *October 2023.*
- **Guest Lecturer:** Guest lecturer in Spatial Planning and Traffic engineering (SPT) program, Aalto university. Data analysis and Visualization using PPGIS. *March 2023.*
- **Guest Lecturer:** Guest lecturer in Urban Studies and Planning (USP) program, Aalto university. How to use GIS in Urban studies and planning. *January 2023.*
- **Guest Lecturer:** Tutor in Urban Studio Challenge 1 (USP) course, Data analysis, GIS, planning and data visualization assistant. *September 2022.*
- **Guest Lecturer:** Guest lecturer in Sustainable Built Environment (SBE) course, Aalto university. GIS, Spatial Thinking and their use cases in sustainability and urban resilience. *November 2021.*
- **Guest Lecturer:** Guest lecturer in Sustainable Built Environment (SBE) course, Aalto university. GIS and its use cases in sustainability and urban resilience. *December 2020.*
- **Guest Lecturer:** Guest lecturer in Digital urban course at Aalto university. Teaching about data analysis and its uses in built environment, types of data and available information. *March 2020.*